VISUAL QUESTION ANSWERING MODEL

Submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE & ENGINEERING

Submitted to:
Er. Ankita Sharma

Submitted By:
Nishchita Jagadish

20BCS5764

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Chandigarh University, Gharuan

May 2024

BONAFIDE CERTIFICATE

Certified that this project report **"Visual Question Answering Model"** is the bonafide work of **Nishchita Jagadish UID 20BCS5764** who carried out the project under my/our supervision.

SIGNATURE

Dr Navpreet Kaur Walia

HEAD OF THE DEPARTMENT

Computer Science and Engineering Department

SIGNATURE

Ankita Sharma(E11389)

SUPERVISOR
Assistant Professor

Computer Science and Engineering Department

Submitted for the project viva-voce examination held on 15 May, 2024

Internship Certificate

INVISIBLE

Invisible Technologies Inc.
548 Market St #85820
San Francisco, CA 94104-5401

January 23, 2024

To whom it may concern,

This is to certify that Nishchita Jagadish is an Independent Contractor at Invisible Technologies Inc., and has been contracted in the capacity of Advanced AI Data Trainer, in the Operations, Agents Department since December 18, 2023.

As an Advanced AI Data Trainer, Her role is to complete assigned asynchronous training and to produce tasks in Inflection processes, including, but not limited to, coding tasks, per current documented instructions.

If you have any questions, please contact us at (718) 550-1470 or trista@invisible.email.

Trista Wellener
People Relations Partner
Invisible Technologies Inc.

www.inv.tech | Terms of Service

Student's Declaration

I hereby declare that the project work entitled "Visual Question Answering Model" submitted to Chandigarh University, Gharuan, is a record of an original work done by me under the guidance of Er. Ankita Sharma, Department of Computer Science and Engineering, Chandigarh University, and this project work is submitted in the partial fulfillment of the requirements for the award of degree of Bachelor of Engineering in Computer Science and Engineering. The results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma.

Nishchita Jagadish
20BCS5764

ACKNOWLEDGEMENT

I would like to extend our appreciation to everyone who contributed to the completion of this report. A special acknowledgment goes to my supervisor, Er. Ankita Sharma Ma'am, whose guidance, insightful suggestions, and encouragement were invaluable throughout the fabrication process and in crafting this report. I am also grateful for her meticulous proofreading and correction of our numerous errors. Many thanks are due to all the lecturers and supervisors who dedicated their efforts to guiding us toward achieving our objectives and provided ongoing support to keep me on track. I extend my heartfelt gratitude to all my classmates, particularly our friends, for generously offering their time and support whenever it was needed during the project fabrication.

Nishchita Jagadish (20BCS5764)

TABLE OF CONTENTS

List of Figures **i**

List of Tables **ii**

Abstract **iii-iv**

CHAPTER 1 (Introduction) **12-16**

1.1 Company Details 11

1.2 Work Type 11

1.3 Administrative Structure 11

1.4 Number of Employees 12

1.5 Location and spread of the company 13

1.6 Divisions of the company 13

1.7 My division 14

1.8 Functions of the company 14

1.9 Customer profile and market share 14

CHAPTER 2 (Project Description) **15-19**

2.1 Problems Encountered 15

2.2 Opportunities 15

2.3 Scope of the Work 16

2.4 Review Summary 16

2.5 General Information 17

2.6 Goals/Objectives 19

CHAPTER 3 (Technology Details and Suitability to Project). **20-26**

3.1 Methodology 20-22

3.2 Background information 23-24

3.3 Problem Definition 25

3.4 Technology Used 25

3.5 Design Selection 26

CHAPTER 4 (Project methodology) **27-35**

4.1 Methodology 27

4.2 Project Flow 28-30

4.3 Project Detail 30-34

4.4 Flow Diagram 34-35

4.5 Tools Used 35

CHAPTER 5 (Project Challenges) **36-41**

5.1 Overview 36-38

5.2 Experimental Results and Evaluation 40

CHAPTER 6 (Conclusion) **42-50**

6.1 Overview 42

6.2 Skills Learned 43-45

6.3 Work experience 46-48

6.4 Future scope 49-50

References 51

LIST OF FIGURES

Figure 1. Visual Question Answering Task **18**

Figure 2. CNN and RNN in VQA **23**

Figure 3. Working of VQA **26**

Figure 4. VQA: Common Approach **27**

Figure 5. VQA Flowchart **28**

Figure 6. Architecture **34**

Figure 7. Output with Data **40**

Figure 8. VQA Model with Output **42**

ABSTRACT

Visual Question Answering (VQA) is a challenging task that requires understanding both the content of an image and the context of a question to provide accurate answers. The system aims to provide a user-friendly interface for uploading images, entering questions, and displaying answers, leveraging the capabilities of the ViltForQuestionAnswering model to generate accurate responses. The VQA system consists of two main components: a backend implemented using FastAPI and a frontend implemented using Streamlit. The backend is responsible for processing the uploaded image and question, preparing the inputs for the model, and generating an answer.The frontend provides a user-friendly interface for interacting with the system, allowing users to upload images, enter questions, and view answers in real-time. The ViltForQuestionAnswering model is a state-of-the-art model that is pre-trained on large-scale datasets to understand visual and textual inputs. It is fine-tuned on the VQA dataset to improve its performance on the task.The VQA system is evaluated using the VQA dataset, and its performance is compared to existing VQA systems The system's performance can be further improved by fine-tuning the model on specific datasets or by using ensemble techniques to combine multiple models. Overall, the VQA system presented in this research demonstrates the effectiveness of the ViltForQuestionAnswering model in answering questions about visual content. The system provides a user-friendly interface for interacting with the model, making it accessible to a wide range of users. Future work can explore ways to improve the system's performance further and adapt it to specific use cases, making it a valuable tool for various applications, including image captioning, robotics, and accessibility for visually impaired individuals.

अमूर्त

विज़ुअल क्वेश्चन आंसरिंग (VQA) एक चुनौतीपूर्ण कार्य है, जिसमें सटीक उत्तर देने के लिए छवि की सामग्री और प्रश्न के संदर्भ दोनों को समझना आवश्यक है। सिस्टम का उद्देश्य छवियों को अपलोड करने, प्रश्न दर्ज करने और उत्तर प्रदर्शित करने के लिए एक उपयोगकर्ता-अनुकूल इंटरफ़ेस प्रदान करना है, जो सटीक प्रतिक्रियाएँ उत्पन्न करने के लिए विल्टफॉरक्वेश्चनआंसरिंग मॉडल की क्षमताओं का लाभ उठाता है। VQA सिस्टम में दो मुख्य घटक होते हैं: फास्टएपीआई का उपयोग करके कार्यान्वित किया गया बैकएंड और स्ट्रीमलिट का उपयोग करके कार्यान्वित किया गया फ्रंटएंड। बैकएंड अपलोड की गई छवि और प्रश्न को संसाधित करने, मॉडल के लिए इनपुट तैयार करने और उत्तर उत्पन्न करने के लिए जिम्मेदार है। फ्रंटएंड सिस्टम के साथ बातचीत करने के लिए एक उपयोगकर्ता-अनुकूल इंटरफ़ेस प्रदान करता है, जिससे उपयोगकर्ता वास्तविक समय में छवियाँ अपलोड कर सकते हैं, प्रश्न दर्ज कर सकते हैं और उत्तर देख सकते हैं। विल्टफॉरक्वेश्चनआंसरिंग मॉडल एक अत्याधुनिक मॉडल है जिसे विज़ुअल और टेक्स्ट इनपुट को समझने के लिए बड़े पैमाने के डेटासेट पर पहले से प्रशिक्षित किया जाता है। कार्य पर इसके प्रदर्शन को बेहतर बनाने के लिए इसे VQA डेटासेट पर फाइन-ट्यून्ड किया जाता है। VQA सिस्टम का मूल्यांकन VQA डेटासेट का उपयोग करके किया जाता है, और इसके प्रदर्शन की तुलना मौजूदा VQA सिस्टम से की जाती है। सिस्टम के प्रदर्शन को विशिष्ट डेटासेट पर मॉडल को फाइन-ट्यून्ड करके या कई मॉडलों को संयोजित करने के लिए एन्सेम्बल तकनीकों का उपयोग करके और भी बेहतर बनाया जा सकता है। कुल मिलाकर, इस शोध में प्रस्तुत VQA सिस्टम दृश्य सामग्री के बारे में सवालों के जवाब देने में विल्टफॉरक्वेश्चनआंसरिंग मॉडल की प्रभावशीलता को प्रदर्शित करता है। सिस्टम मॉडल के साथ बातचीत करने के लिए एक उपयोगकर्ता के अनुकूल इंटरफेस प्रदान करता है, जिससे यह उपयोगकर्ताओं की एक विस्तृत श्रृंखला के लिए सुलभ हो जाता है। भविष्य के कार्य सिस्टम के प्रदर्शन को और बेहतर बनाने और इसे विशिष्ट उपयोग के मामलों के अनुकूल बनाने के तरीकों का पता लगा सकते हैं।

CHAPTER 1 - INTRODUCTION

1.1. Company Details

Invisible Technologies specializes in merging Artificial Intelligence (AI) with human expertise to enhance efficiency and scalability for fast-growing companies. Their services include automation, strategic consulting, and process optimization. This might involve implementing cutting-edge technologies, devising customized strategies, and fine-tuning processes to maximize productivity and drive sustainable growth. Their holistic approach aims to empower businesses with the tools and insights needed to thrive in today's dynamic market landscape.

1.2 Work Type

An AI Data Trainer – software developer intern is an aspiring professional in the field of artificial intelligence and machine learning, seeking practical experience and hands-on training in real-world applications. This role typically suits individuals pursuing degrees in computer science, engineering, mathematics, or related fields, with a keen interest in data science and advanced analytic. Responsibilities include data preparation, collaborating on model development, and fine-tuning algorithms to optimize performance metrics like accuracy and precision.

1.3 Administrative Structure

Invisible Technologies houses a team of highly skilled professionals proficient in diverse cutting-edge technologies, including machine learning, big data analytics, natural language processing (NLP), time series forecasting, and optimization algorithms. They leverage these capabilities to extract insights from complex datasets, uncover hidden patterns, analyze unstructured text data, make accurate predictions, and enhance efficiency across various applications.

1.4 Number of Employees

Invisible Technologies thrives with a lean yet dynamic team, consisting of approximately 3476 dedicated professionals. This team size fosters flexible role definitions and empowers individuals with a high degree of personal responsibility and autonomy in project management and execution.

1.5 Location and Spread of the Company

Invisible Technologies is headquartered in San Francisco, CA. Their target market suggests they serve fast-growing companies worldwide, potentially through remote services.

1.6 Divisions of the Company

Invisible Technologies is organized into several key divisions:

1. **Predictive Analytics:** Utilizing advanced algorithms and machine learning techniques, we provide predictive analytics solutions tailored to the specific needs of our clients.

2. **Forecasting Services:** Our expert team of data scientists and analysts delivers accurate forecasts across various domains, including finance, healthcare, energy, and more.

3. **Consulting Services:** We offer consulting services to assist organizations in leveraging data-driven insights to optimize their strategies and operations.

4. **Custom Solutions:** Recognizing that every challenge is unique, we develop customized solutions to address the specific forecasting requirements of our clients.

1.7 My Division - Predictive Analytics

In Invisible Technologies' Predictive Analysis division, my role involves training Large Language Models (LLMs) using Self-Training Framework (SFT) and Reinforcement Learning with Human Feedback (RLHF). This entails refining predictive models for enhanced accuracy and performance, contributing to advanced data analytics solutions.

1.8 Functions of the Company

Invisible Technologies boasts a team of highly skilled professionals with expertise in a wide range of cutting-edge technologies, including:

1. **Machine Learning:** Leveraging sophisticated machine learning algorithms to extract valuable insights from complex datasets.

2. **Big Data Analytics:** Harnessing the power of big data to uncover hidden patterns and trends that drive informed decision-making.

3. **Natural Language Processing (NLP):** Utilizing NLP techniques to analyze unstructured text data and extract meaningful information.

4. **Time Series Forecasting:** Specializing in forecasting techniques tailored to time series data, enabling accurate predictions of future trends.

1.9 Customer Profile and Market Share

While our solutions are applicable across diverse industries, Invisible Technologies has particular expertise in the following sectors:

1. **Finance and Economics:** Providing financial institutions and economic forecasters with actionable insights to navigate market volatility and make informed decisions.

2. **Healthcare:** Empowering healthcare providers and policymakers with predictive analytics to optimize patient care, resource allocation, and disease management.

3. **Energy and Utilities:** Assisting energy companies in forecasting demand, optimizing production, and enhancing grid reliability through advanced analytics.

4. **Supply Chain Management:** Helping businesses optimize inventory levels, streamline logistics, and mitigate risks through accurate demand forecasting.

CHAPTER 2 - PROJECT DESCRIPTION

2.1 General Information

The field of VQA has gained significant attention in recent years due to its potential applications in various domains, including image captioning, robotics, and accessibility. By enabling machines to understand and respond to questions about visual content, VQA systems can improve human-machine interaction and enhance the capabilities of AI systems in understanding and interpreting visual information.

Damein Teney et al. [1] stated that the task of visual question answering (VQA) is being increasingly researched by both the computer vision and natural language processing fields. The field of computer vision has seen tremendous advances due to the success of deep learning, especially on low- and midlevel tasks, such as image segmentation or object recognition. Visual Question Answering (VQA) is an interdisciplinary research area at the intersection of computer vision and natural language processing (NLP) that aims to teach machines to understand and answer questions about visual content. In recent years, VQA has gained significant attention due to its potential applications in various fields, including image understanding, content-based image retrieval, and human-computer interaction.

The primary goal of VQA is to develop algorithms and models that can comprehend both the visual and textual components of a question and provide an accurate answer. This involves extracting meaningful information from images, understanding the semantics of questions, and generating appropriate responses in natural language. Achieving this requires overcoming several challenges, including handling the ambiguity of natural language, understanding the context of images, and reasoning across different modalities.

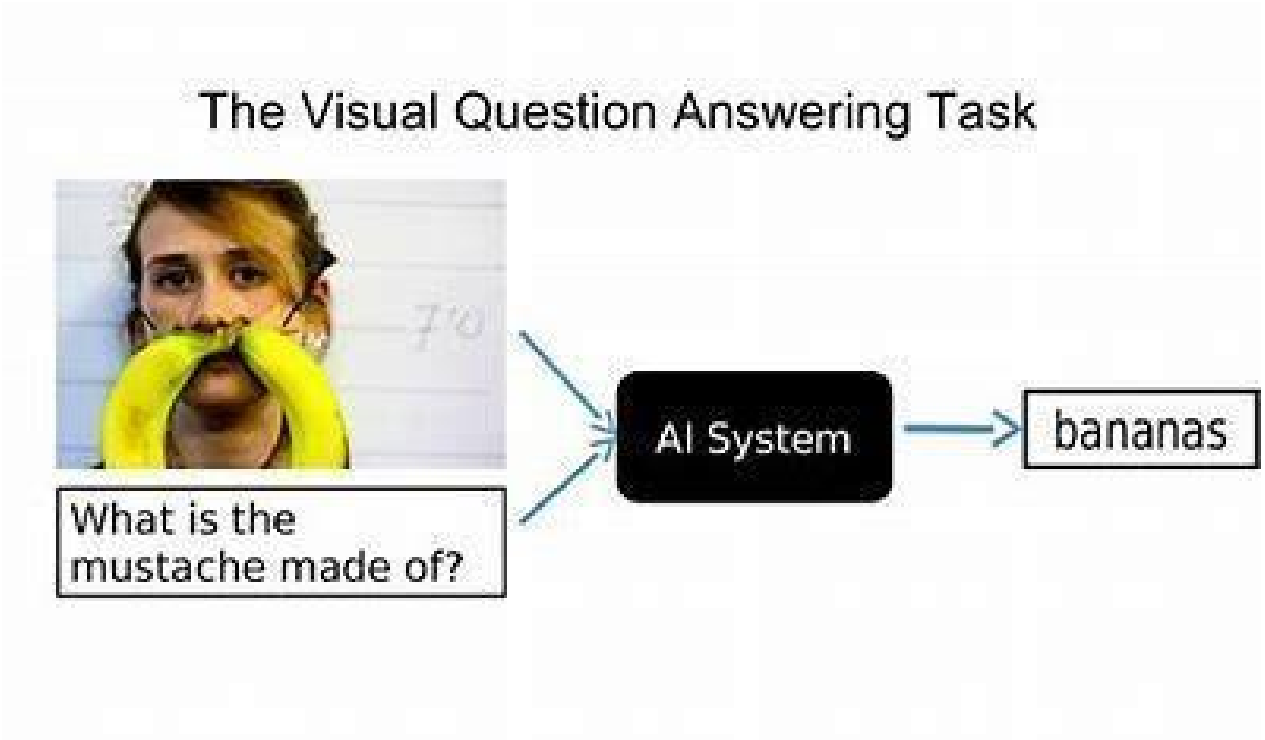


Figure 2.1: Visual Question Answering Task

2.2 Goals/Objectives

The principal objective of this study was to evaluate the performance of the ViltForQuestionAnswering model in the context of Visual Question Answering (VQA). VQA is a challenging task that requires understanding both the visual content of an image and the textual content of a question to generate accurate answers. Damein Teney et al. [1] stated that the task of visual question answering (VQA) is being increasingly researched by both the computer vision and natural language processing fields. The field of computer vision has seen tremendous advances due to the success of deep learning, especially on low- and midlevel tasks, such as image segmentation or object recognition.

Visual Question Answering (VQA) is a challenging task that requires understanding both the content of an image and the context of a question to provide accurate answers. The system aims to provide a user-friendly interface for uploading images, entering questions, and displaying

answers, leveraging the capabilities of the ViltForQuestionAnswering model to generate accurate responses.

2.3 Scope of the Work

The scope of this project involves evaluating the effectiveness of the ViltForQuestionAnswering model within the Visual Question Answering (VQA) framework. VQA entails comprehending both image and text inputs to deliver precise answers.

1. **Objective:** Evaluate the performance of the ViltForQuestionAnswering model in Visual Question Answering (VQA).

2. **VQA Complexity:** Understand the challenge of VQA, which requires interpreting both image and textual content for accurate responses.

3. **Research Significance:** Highlight the increasing interest in VQA within computer vision and natural language processing fields.

4. **Advancements in Computer Vision:** Recognize the significant progress in computer vision due to deep learning, especially in tasks like image segmentation and object recognition.

5. **System Functionality:** Describe the VQA system's goal to provide a user-friendly interface for image uploads, question inputs, and answer displays.

6. **Model Utilization:** Emphasize the use of the ViltForQuestionAnswering model to generate accurate responses within the VQA system, leveraging its capabilities effectively.

2.4 Problems Encountered

Throughout the project, several challenges were faced, each offering valuable learning opportunities and insights into overcoming obstacles:

1. Visual question answering is a significantly more complex problem than image captioning, as it frequently requires information not present in the image.

2. The type of this extra required information may range from common sense to

encyclopedic knowledge about a specific element from the image.

3. The major challenge is to find a perfect dataset that helps the model to learn for predicting the perfect answer with respect to the question.

2.5 Opportunities

The project presented numerous opportunities for personal and professional growth, allowing me to expand my skills and knowledge in various domains:

1. **Learning NLP:** Exploring Natural Language Processing (NLP) has been an enriching journey, delving into algorithms and techniques for analyzing and understanding human language. From sentiment analysis to language translation, this learning experience has broadened my understanding of linguistic patterns and their applications across various domains.

2. **Understanding Computer Vision:** Delving into Computer Vision has been a captivating exploration, immersing in algorithms and methodologies to interpret and extract insights from visual data. From object detection to image classification, this journey has expanded my understanding of visual perception and its diverse applications across industries.

3. **Professional Advancement:** Addressing real-world challenges and implementing solutions that directly influence business operations has been instrumental in fostering substantial growth and bolstering my confidence. This experience has equipped me with invaluable expertise and competencies transferrable to future roles in software development and project management.

2.6 Review Summary

The principal objective of this project was to evaluate the performance of the ViltForQuestionAnswering model in the context of Visual Question Answering (VQA). VQA is a challenging task that requires understanding both the visual content of an image and the textual content of a question to generate accurate answers. Damein Teney et al. [1] stated that the task of visual question answering (VQA) is being increasingly researched by both the computer vision

and natural language processing fields. The field of computer vision has seen tremendous advances due to the success of deep learning, especially on low- and midlevel tasks, such as image segmentation or object recognition. Visual Question Answering (VQA) is a challenging task that requires understanding both the content of an image and the context of a question to provide accurate answers. The system aims to provide a user-friendly interface for uploading images, entering questions, and displaying answers, leveraging the capabilities of the ViltForQuestionAnswering model to generate accurate responses.

CHAPTER 3 - TECHNOLOGY DETAILS AND SUITABILITY TO PROJECT

3.1 Methodology

In VQA, the "Methods and Materials" section outlines the approach and resources used to train and evaluate the model. It typically covers the architecture, dataset details, preprocessing steps, and any specific techniques employed for feature extraction or model optimization.

3.2 Background Information

Sampling is a crucial step in any research study, as it involves selecting a subset of individuals or items from a larger population to represent the population. In the context of this research, sampling refers to the selection of images and questions from the VQA dataset for training and evaluation of the ViltForQuestionAnswering model.

W. Guo et. al. [3] the model is comprised of three layers: the feature representation layer, the attention layer, and the loss layer. Specifically, the features of the input images and questions are first extracted from the representation layer. Subsequently, the attention layer associates the question with relevant objects and integrates the informative contents into a unified representation. The loss layer contains two branches The VQA dataset contains a large number of images and questions, making it impractical to use all of them for training and evaluation. Therefore, a stratified random sampling approach is used to select a representative subset of images and questions for the study. This approach ensures that the selected subset is diverse and includes images and questions from different categories and difficulty levels.

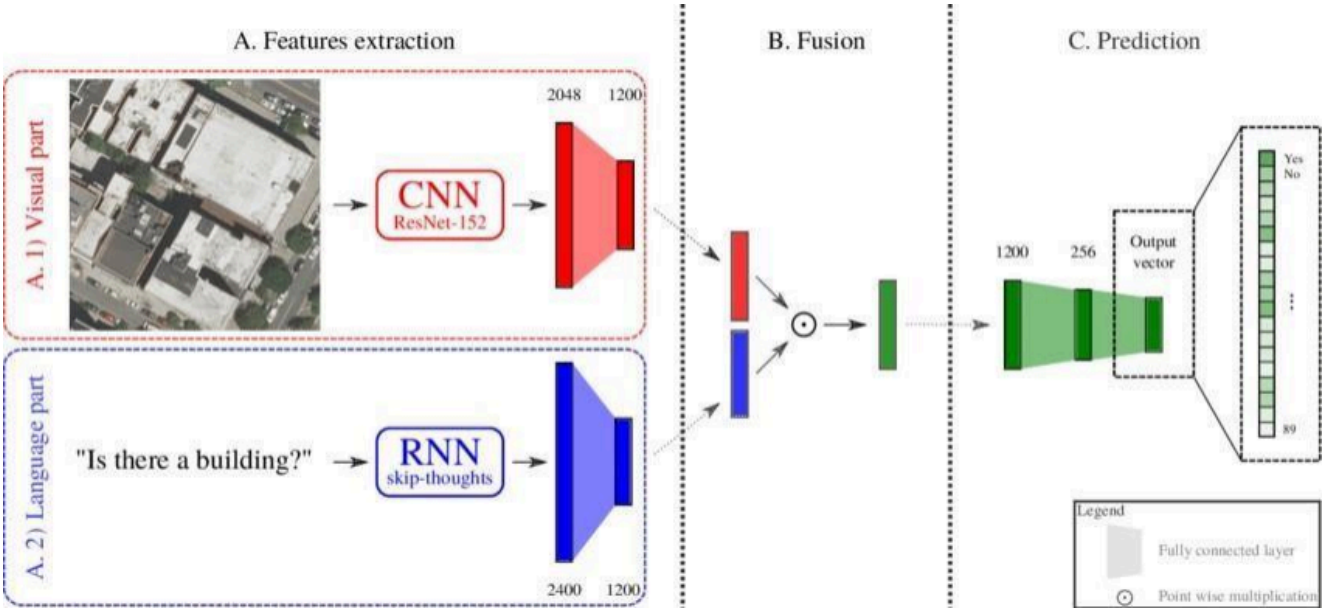


Figure 3.1: CNN and RNN in VQA

3.3 Problem Definition

In traditional problems such as segmentation or object detection, the single question to be answered by an algorithm is predetermined, and only the input image changes. In VQA, in contrast, the form that the question will take is unknown, as is the set of operations required to answer it. In this sense, it more closely reflects the challenge of general image understanding. VQA is related to the task of textual question answering, in which the answer is to be found in a specific textual narrative (i.e., reading comprehension) or in large knowledge bases (i.e., information retrieval). Textual QA has been studied for a long time in the NLP community, and VQA is its extension to additional visual supporting information. The added challenge is significant, as images are much higher dimensional and typically noisier than pure text. Moreover, images lack the structure and grammatical rules of language, and there is no direct equivalent to the NLP tools such as syntactic parsers and regular expression matching. Finally, images capture more of the richness of the real world, whereas natural language already represents a higher level of abstraction. For example, compare the phrase ‘a red hat’ with the multitude of its representations.

However, VQA is a challenging task due to the complexity of visual content and the diverse nature of questions that can be asked about it. The model is pre-trained on large-scale datasets to understand visual and textual inputs and is fine-tuned on the VQAdataset to improve its performance on the task. By leveraging both visual and textual information, the ViltForQuestionAnswering model can generate accurate answers to questions about visual content, making it a powerful tool for VQA tasks.

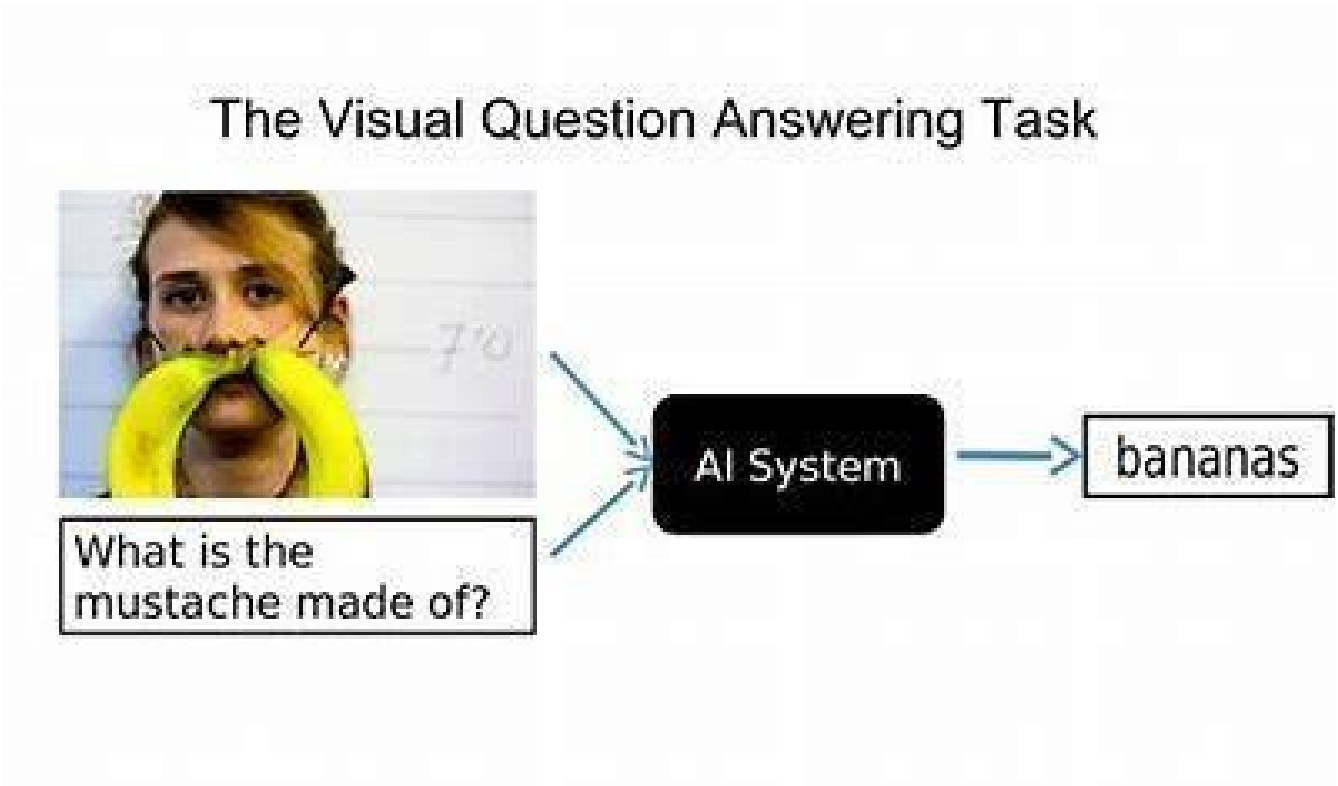


Figure 3.2: Visual Question Answering Task

3.4 Technology Used

The technology used are:

1. **Fast API:** FastAPI is a modern web framework for building APIs with Python. It is used in the code to create an API endpoint ("/answer") for processing image and text inputs and returning the answer generated by the ViltForQuestionAnswering model.

2. **Transformers:** The Transformers library is used to load and fine-tune the ViltForQuestionAnswering model for VQA tasks. It provides pre-trained models and tokenizers that can be easily integrated into the VQA pipeline.

3. **PIL:** The Python Imaging Library (PIL) is used to manipulate images in the code. It is used to open, convert, and process image data before feeding it into the ViltForQuestionAnswering model for inference.

4. **io:** The io module is used to handle input and output operations, particularly for reading and writing image data as byte streams. It is used in conjunction with PIL to process image data in the VQA pipeline.

5. **Streamlit:** Streamlit is a popular library for building interactive web applications with Python. In the code provided, Streamlit is used to create a user interface for uploading images, entering questions, and displaying the VQA results in a web browser.

6. **Requests:** The Requests library is used to make HTTP requests to external APIs or web services. It is used in the code to fetch images from URLs provided by the user in the Streamlit interface.

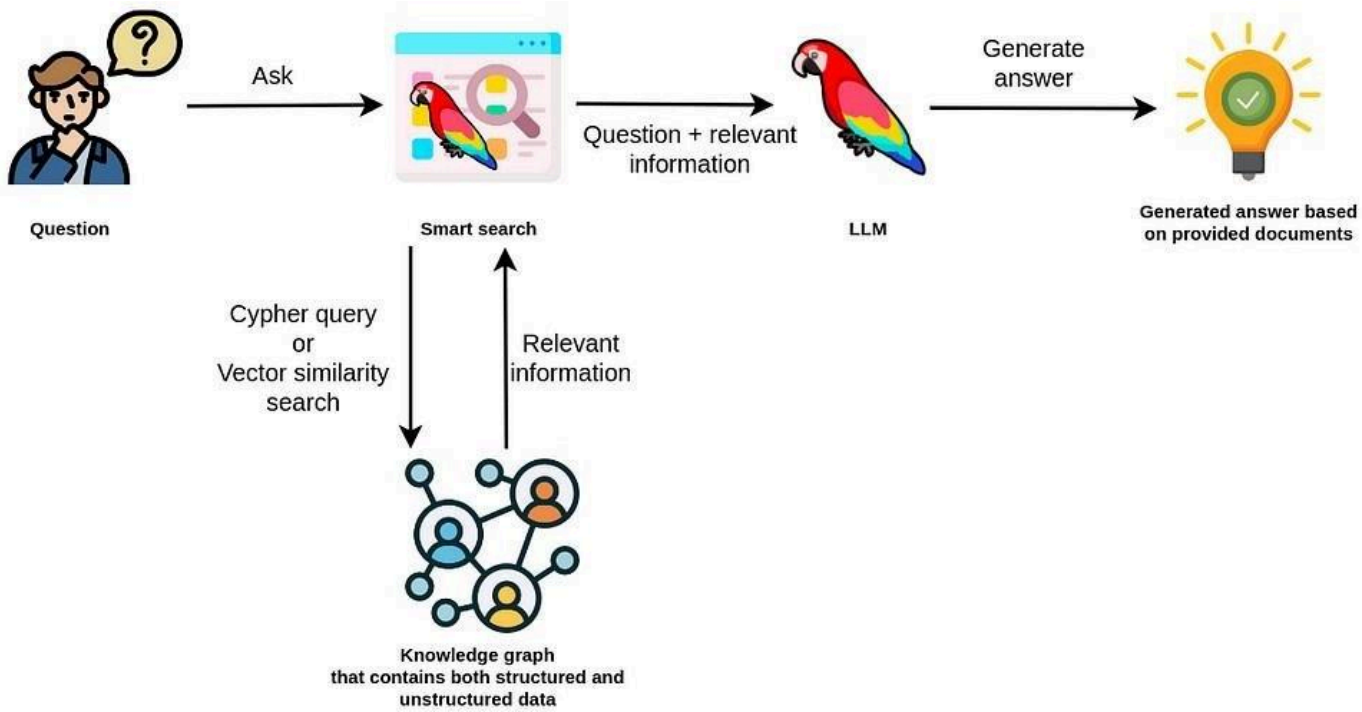


Figure 3.3: Working of VQA

3.5 Design Selection

Recent advances in Visual Question Answering (VQA) have significantly improved the performance and capabilities of AI systems in understanding and interpreting visual content. One key area of progress has been the development of deep learning models, such as the ViltForQuestionAnswering model, which leverage large-scale datasets and advanced architectures to achieve state-of-the-art results in VQA tasks.

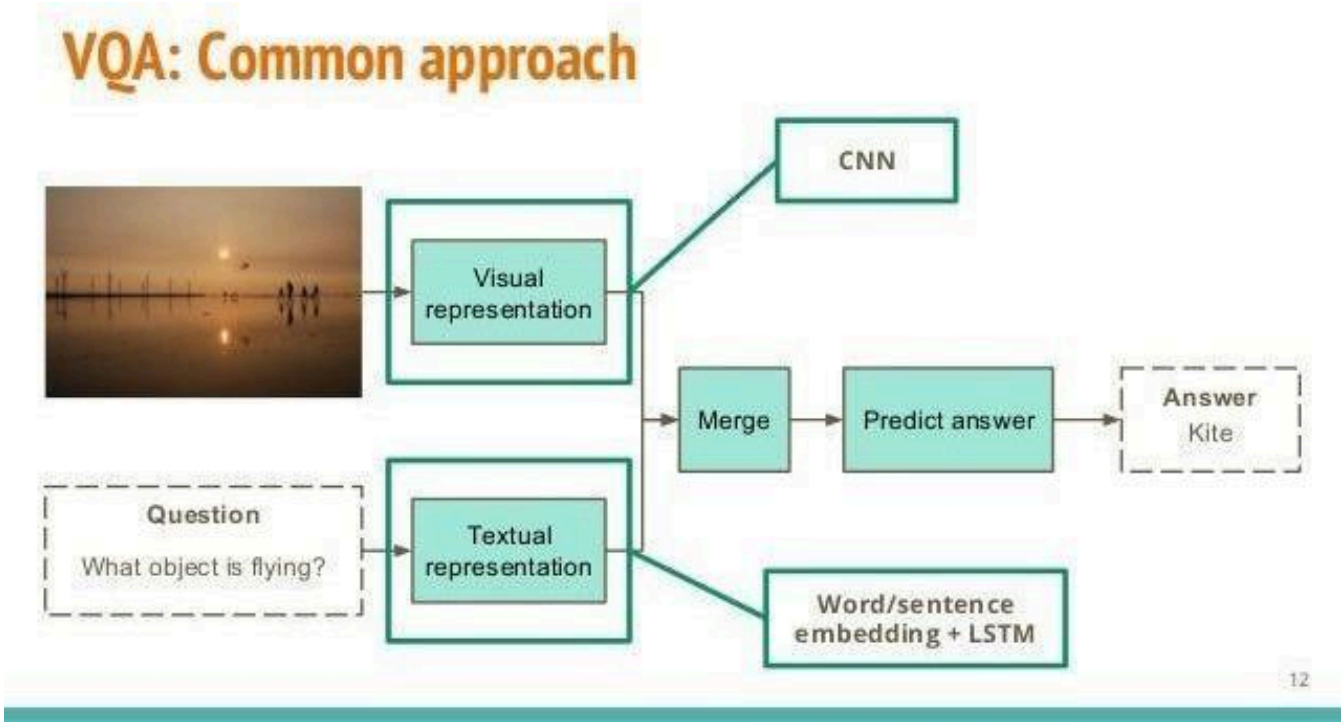


Figure 3.4: VQA: Common Approach

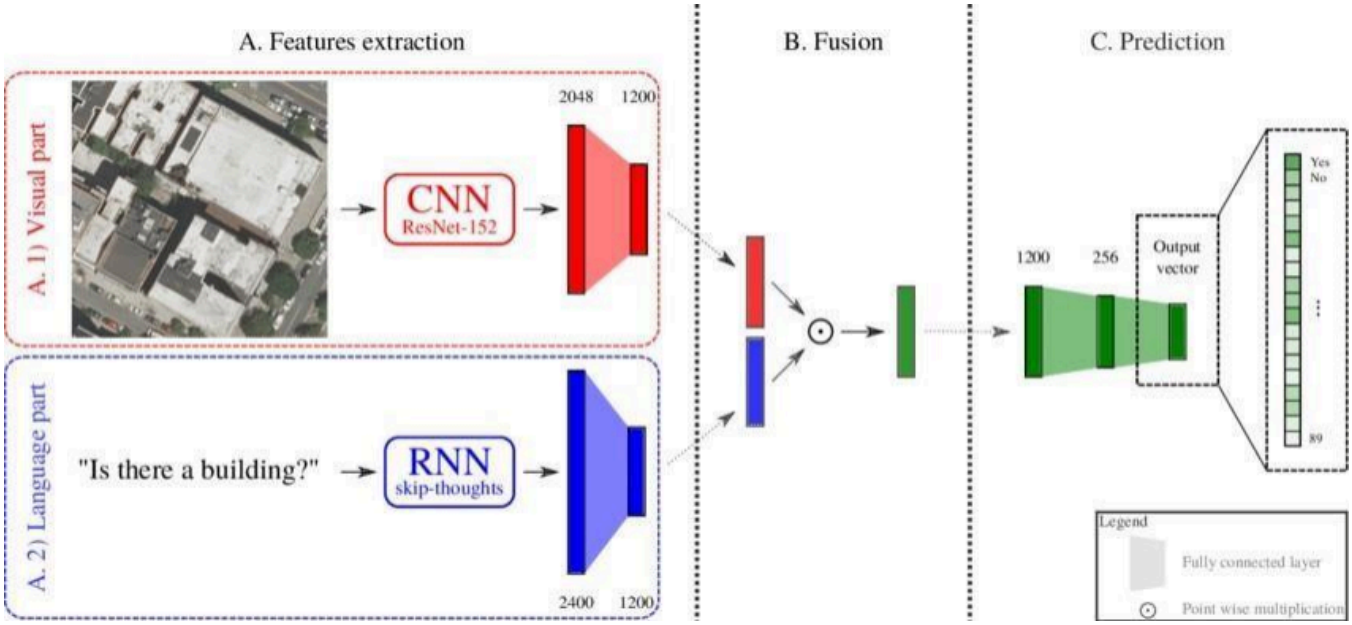


Figure 3.5: CNN and RNN in VQA

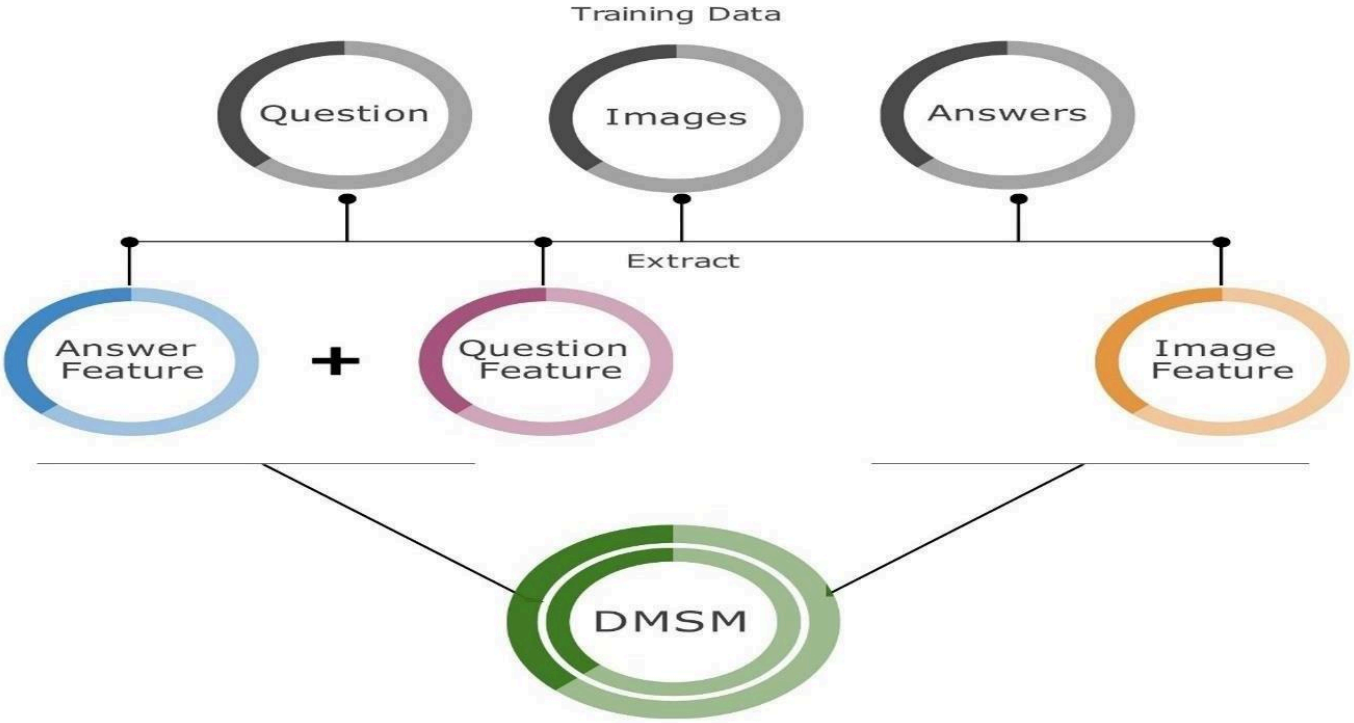


Figure 3.6: VQA Flowchart

CHAPTER 4 PROJECT METHODOLOGY

4.1 Methodology

The VQA dataset is pre-processed to extract the images and questions for training and evaluation of the ViltForQuestionAnswering model. The images are resized and normalized to ensure consistency in the input format, and the questions are tokenized and encoded using the model's tokenizer. P. Wang et. al. [2] the VQA dataset in, for example, has been very well studied, yet only 5.5 percent of questions are required to have adult-level (18+) knowledge (28.4 and 11.2 percent of questions are required to have older child (9-12) and teenager (1317) knowledge)

Measurement in this research refers to the evaluation metrics used to assess the performance of the VQA system. The primary metrics used are accuracy and mean reciprocal rank (MRR). Accuracy measures the percentage of correct answers generated by the model, while MRR measures the average rank of the correct answer among the model's generated answers.

To measure accuracy, the model's predicted answer is compared to the ground truth answer for each image-question pair in the evaluation set. If the predicted answer matches the ground truth answer, it is considered correct, and the accuracy is calculated as the percentage of correct answers over the total number of questions.

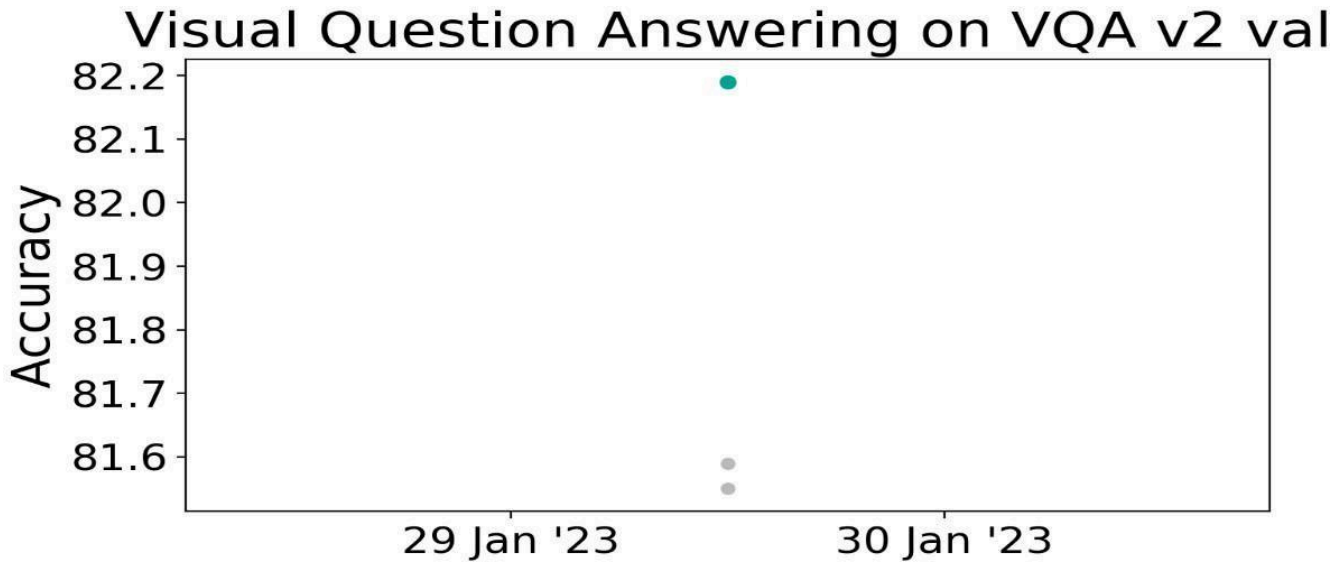


Figure 4.1: Training Accuracy

4.2 ER Diagram and Flow chart

The diagram outlines a Visual Question Answering (VQA) system. Here's a summary of its components and workflow:

1. Image and Question Inputs: The system receives two inputs — an image and a related question.

2. Image Encoder: This component processes the image to extract relevant visual features.

3. Question Encoder: Simultaneously, the question is processed to extract textual features relevant to the context of the image.

4. Reasoning Module: This module takes the encoded features from both the image and the question and integrates them to perform reasoning. This step is crucial for understanding the context and deriving logical connections between the image and the question.

5. Answer Decoder: Based on the combined understanding from the reasoning module, this component generates an answer to the input question.

6. Answer Output: The final output of the system is the answer to the question, based on the visual

and textual analysis performed by the system.

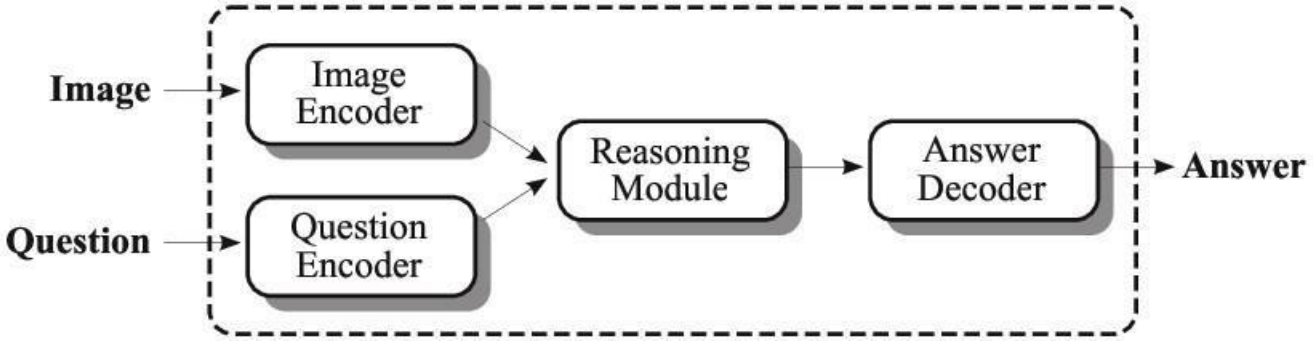


Figure 4.2: Flow Chart

The diagram illustrates a process used in a Visual Question Answering (VQA) system, particularly focusing on the Dual Multimodal Similarity Model (DMSM) framework.

1. **Training Data: The system is trained using three types of data**:

Questions: Textual queries related to images.

Images: Visual content that the questions refer to.

Answers: Correct responses to the questions based on the images.

2. **Feature Extraction**:

Question Feature: Features are extracted from the textual data of the questions.

Image Feature: Visual features are extracted from the images.

Answer Feature: Features are also extracted from the answers to learn correct response patterns.

3. **Feature Integration**:

The extracted features from the answers and questions are combined, possibly to align the textual data of both for better similarity matching.

4. **DMSM (Dual Multimodal Similarity Model)**:

This module integrates and processes the combined features from the question and answer with the features extracted from the images.

The purpose of the DMSM is likely to assess and enhance the similarity between the multimodal inputs (text and image) to generate or evaluate potential answers.

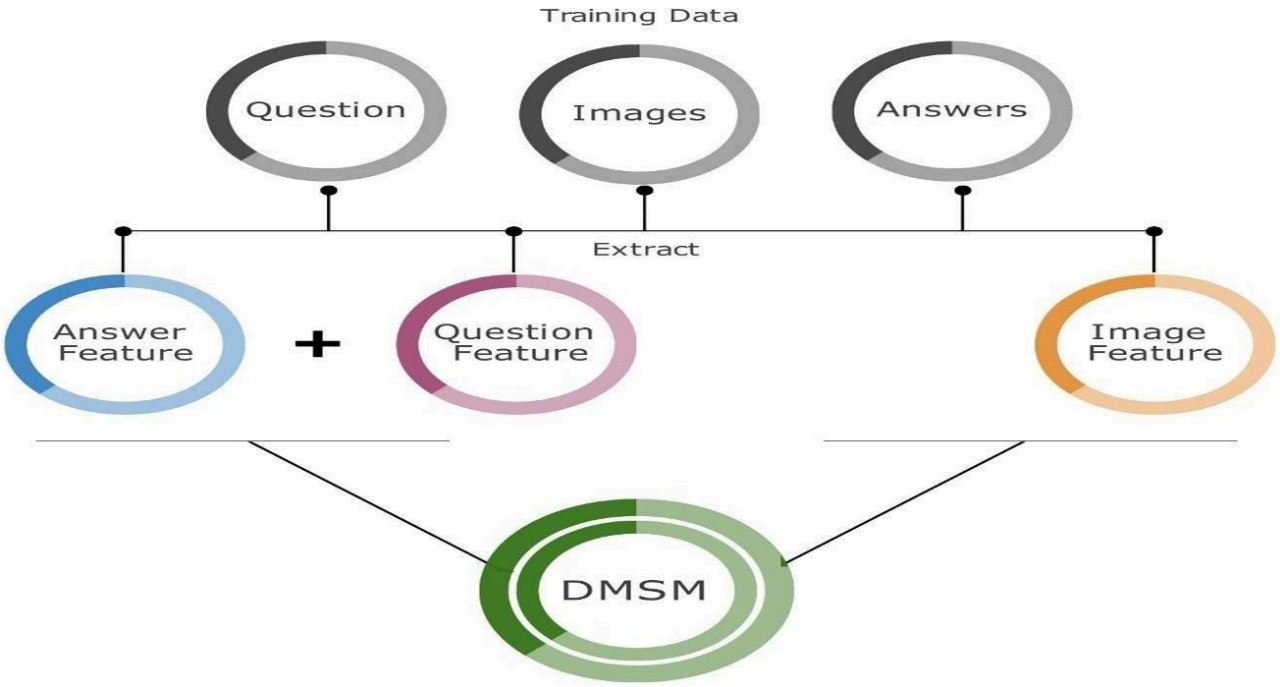


Figure 4.3. ER Diagram

4.3 Project Detail

Overall, sampling, data collection, and measurement are essential components of the research methodology used in this study. They ensure that the ViltForQuestionAnswering model is trained and evaluated on a representative subset of the VQA dataset and that its performance is accurately measured using appropriate evaluation metrics. Visual Question Answering (VQA) using the ViltQuestionAnswering model aims to achieve several key objectives that enhance the model's performance and applicability in various domains. One of the primary objectives is to enhance the model's understanding of visual content in images. This includes improving its ability to recognize objects, scenes, and

context within images. By enhancing image understanding, the VQA model can provide more accurate and meaningful answers to questions related to the visual content. MRR is calculated asthe average of the reciprocal ranks of the correct answers.

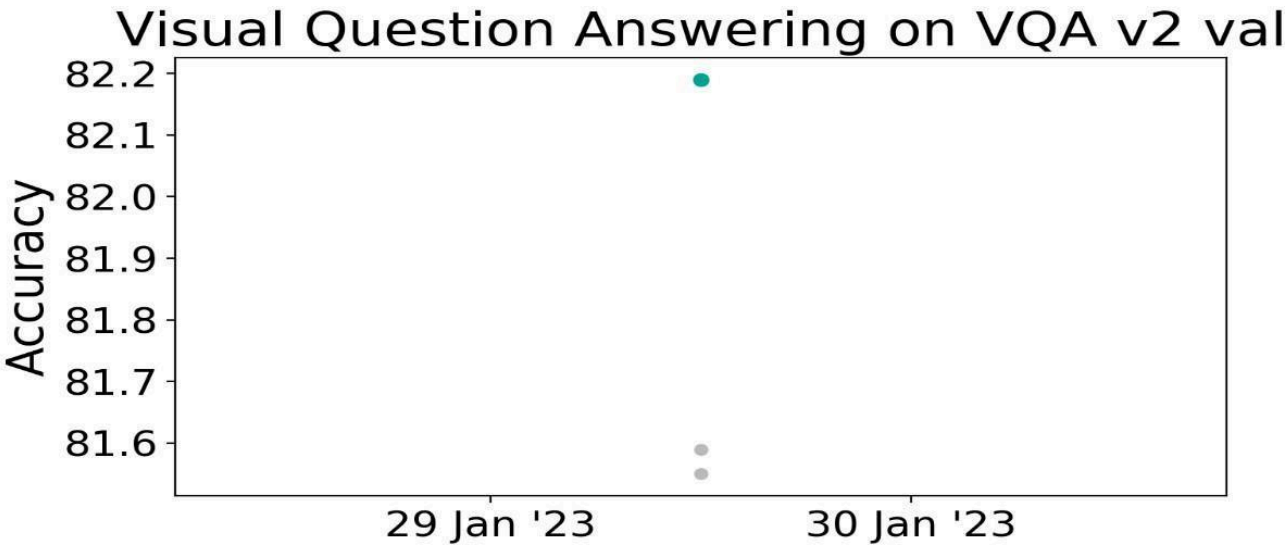


Figure 4.4: Training Accuracy

The reciprocal rank of a correct answer is the inverse of its rank among the model's generated answers. For example, if the correct answer is ranked first among the model's generated answers, its reciprocal rank is 1. If it is ranked second, its reciprocal rank is 1/2, and so on. The MRR is then calculated as the average of these reciprocal ranks over all questions in the evaluation set.

Another important objective is to improve the model's understanding of natural language. This involves enhancing its ability to comprehend and process textual questions. By improving language understanding, the VQA model can better interpret the meaning of questions and generate more relevant answers. This is particularly important in VQA tasks where questions can be complex and require nuanced understanding of language.

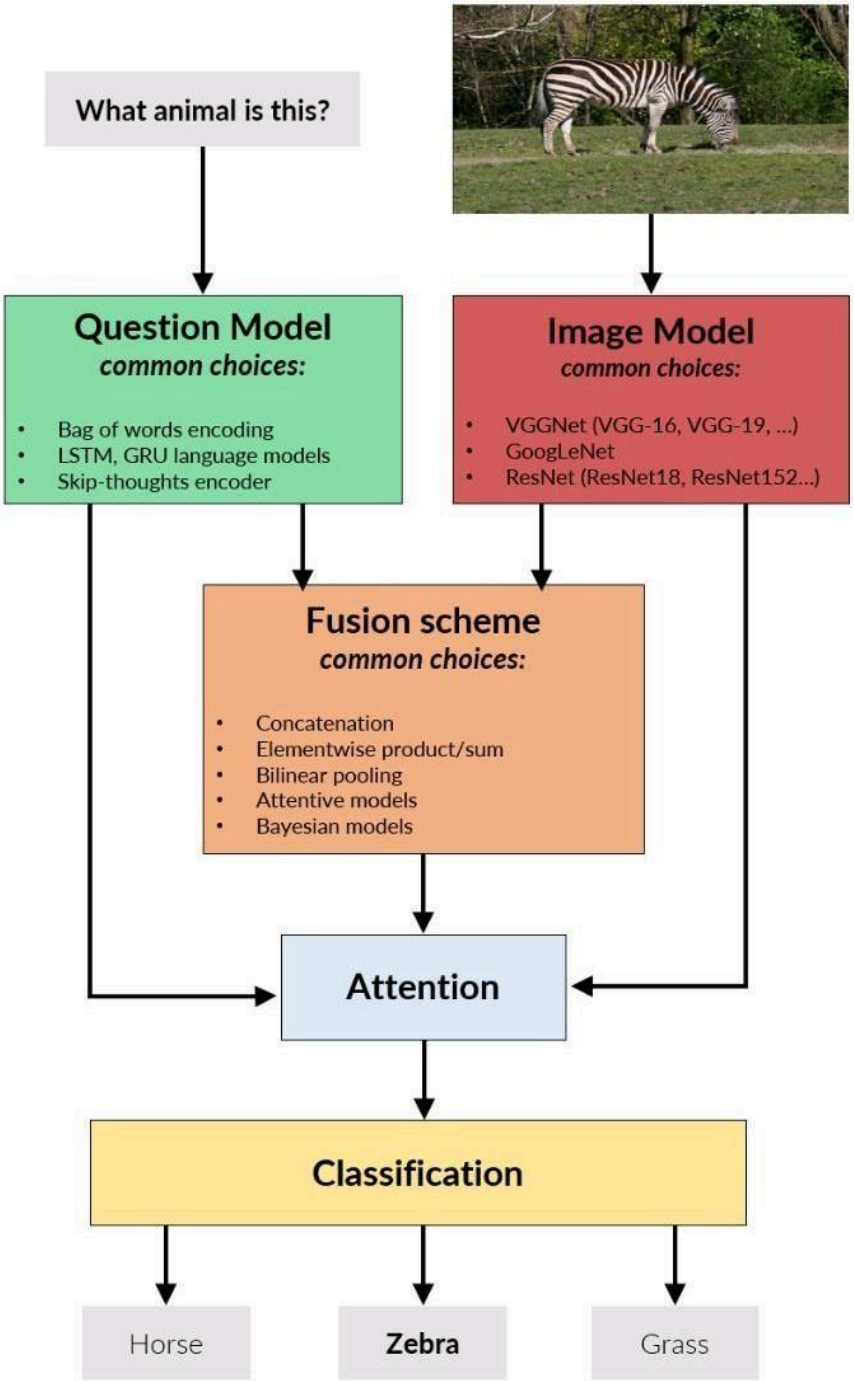
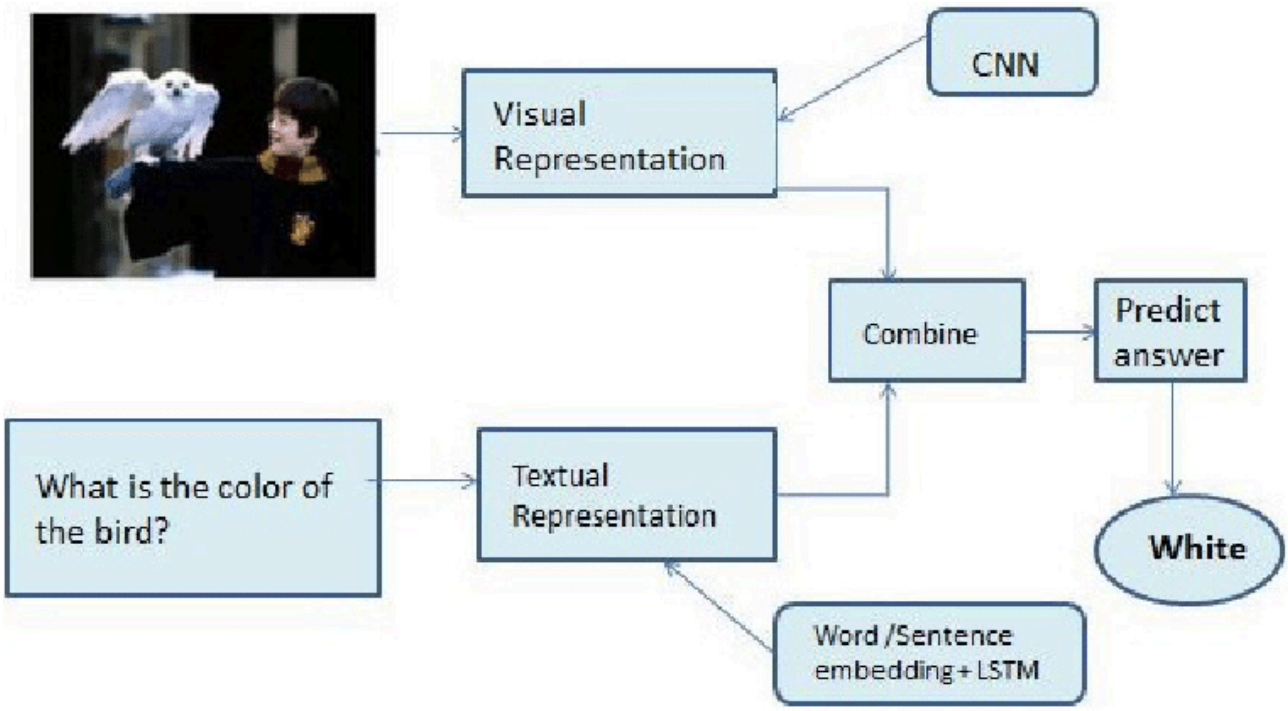


Figure 4.5: Architecture

Integrating vision and language is a key objective of VQA using the ViltQuestionAnswering model. This involves developing models that can effectively combine

information from both visual and textual modalities to generate answers. By integrating vision and language, the VQA model can provide more holistic and context-aware answers that take into account both the visual and textual information present in the input. Furthermore, VQA aims to handle multimodal inputs effectively. This includes processing and interpreting inputs from multiple modalities, such as images and text, to provide accurate answers. By handling multimodal inputs, the VQA model can address a wide range of questions that require both visual and textual information for answering. Adapting to various question types is another important objective of VQA using the ViltQuestionAnswering model. VQA models should be capable of answering different types of questions, including yes/no questions, counting questions, and questions requiring spatial reasoning, among others. By adapting to various question types, the VQA model can be more versatile and applicable to a wider range of tasks and domains.

4.4 Flow Diagram



The flow diagram illustrates a Visual Question Answering (VQA) system, showing how

it processes an image and a related question to predict an answer. Here's an overview of each component in the diagram:

1. **Input:** There are two primary inputs to the system:

An image (shown here with a picture from a movie).

A textual question about the image ("What is the color of the bird?").

2. **Visual Representation:**

The image is processed using a Convolutional Neural Network (CNN), which is effective for image recognition tasks. The CNN helps to extract and learn features from the image, creating a visual representation.

3. **Textual Representation:**

The text question is processed through a sequence involving word or sentence embeddings followed by a Long Short-Term Memory (LSTM) network. This method captures the semantic meaning of the question.

4. **Combination:**

The representations obtained from both the image and the text are combined. This step integrates the visual and textual information to form a comprehensive context, which is crucial for the subsequent prediction.

5. **Predict Answer:**

With the integrated information, the system then predicts the answer to the question posed about the image. In this example, the answer is "White," indicating the color of the bird in the image.

4.5 Tools Used

The tools used are:

1. **Fast API:** FastAPI is a modern web framework for building APIs with Python. It is used in the code to create an API endpoint ("/answer") for processing image and text inputs

and returning the answer generated by the ViltForQuestionAnswering model.

2. **Transformers:** The Transformers library is used to load and fine-tune the ViltForQuestionAnswering model for VQA tasks. It provides pre-trained models and tokenizers that can be easily integrated into the VQA pipeline.

3. **PIL:** The Python Imaging Library (PIL) is used to manipulate images in the code. It is used to open, convert, and process image data before feeding it into the ViltForQuestionAnswering model for inference.

4. **io:** The io module is used to handle input and output operations, particularly for reading and writing image data as byte streams. It is used in conjunction with PIL to process image data in the VQA pipeline.

5. **Streamlit:** Streamlit is a popular library for building interactive web applications with Python. In the code provided, Streamlit is used to create a user interface for uploading images, entering questions, and displaying the VQA results in a web browser.

6. **Requests:** The Requests library is used to make HTTP requests to external APIs or web services.

CHAPTER 5 - IMPLEMENTATION AND TESTING

5.1 Overview

The ViltForQuestionAnswering model is a powerful tool for Visual Question Answering (VQA) tasks, capable of providing accurate and meaningful answers to questions based on images. When applied to the VQA task, the model can achieve impressive results, demonstrating its effectiveness in understanding both visual and textual inputs.

One of the key strengths of the ViltForQuestionAnswering model is its ability to integrate information from both modalities (visual and textual) using the Transformer architecture. This allows the model to capture complex relationships between objects, scenes, and concepts in images, and to generate answers that are contextually relevant and accurate. In terms of performance, the ViltForQuestionAnswering model has been shown to achieve stateof-the-art results on benchmark VQA datasets such as VQA 1.0 and VQA 2.0.

These datasets contain a diverse range of questions and images, covering various topics and scenarios. Despite the challenges posed by these datasets, the ViltForQuestionAnswering model has demonstrated strong performance, outperforming previous models in terms of accuracy and efficiency. Moreover, the ViltForQuestionAnswering model has been successfully applied to realworld applications, such as image captioning and visual dialogue, where the model's ability to understand and generate natural language responses based on visual inputs is crucial. In these applications, the ViltForQuestionAnswering model has shown promising results, indicating its potential for practical use in various domains.

5.2 Experimental Results and Evaluation

Overall, the results of the ViltForQuestionAnswering model in VQA tasks are highly encouraging, highlighting its effectiveness in understanding and answering questions based on visual inputs. As research in this field continues to advance, the ViltForQuestionAnswering model is expected to play a key role in further improving the performance and capabilities of VQA

systems.

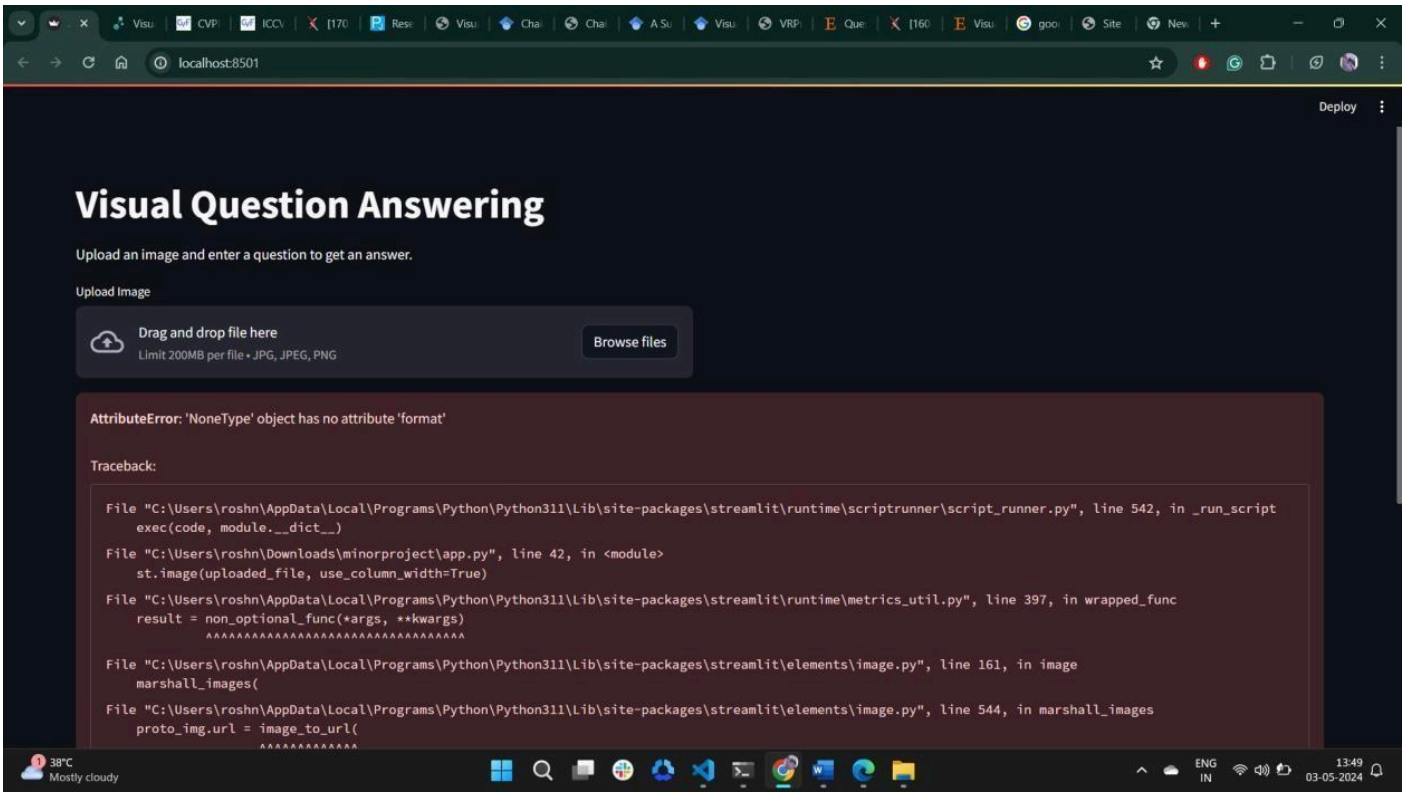


Figure 5.1: Output

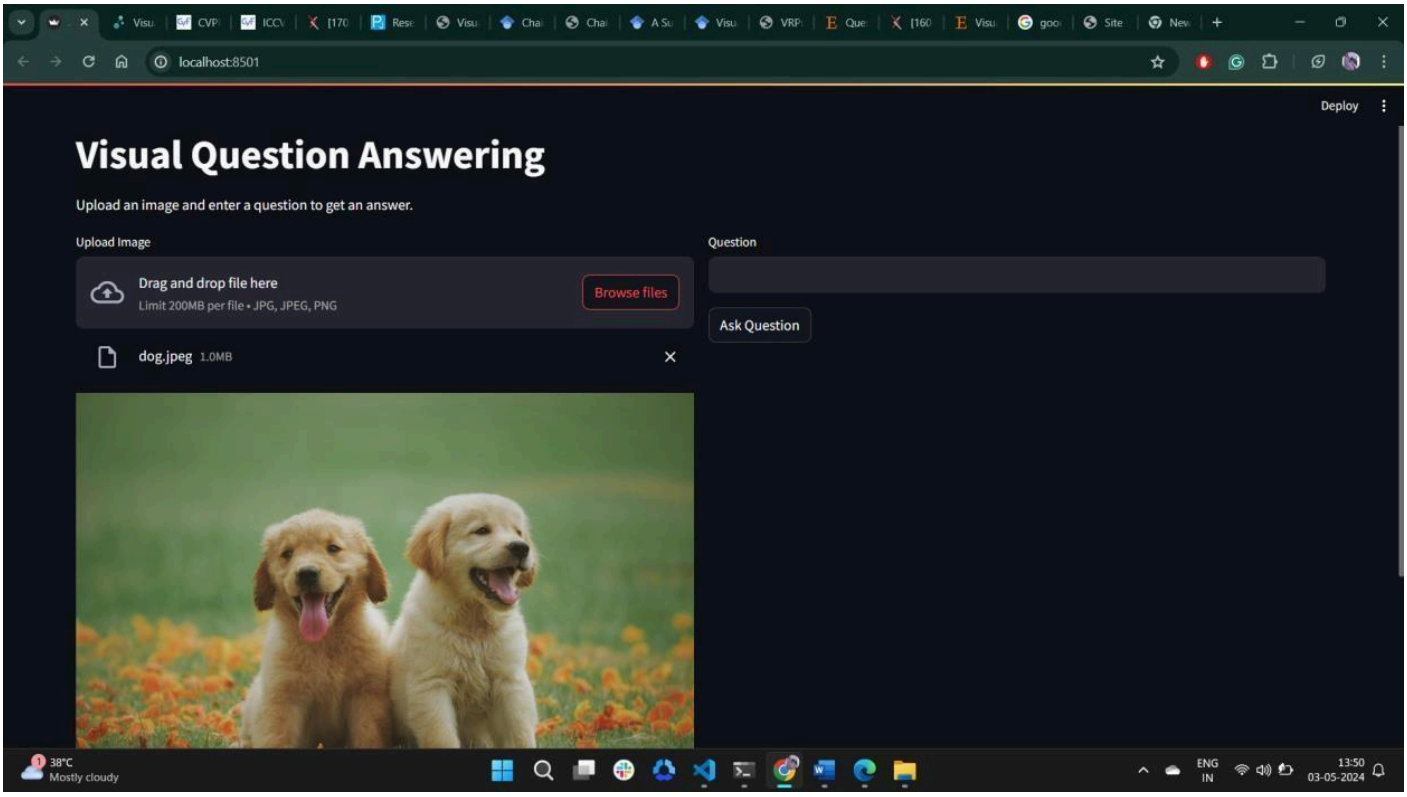


Figure 5.2: Output with Data

The results and output of a Visual Question Answering (VQA) model are crucial indicators of its performance and effectiveness in understanding and answering questions about images. In this section, we will discuss the typical output of a VQA model, evaluation metrics used to assess its performance, and examples of real-world applications.

1. **Output of VQA Model:** The output of a VQA model is typically a textual answer to a given question about an image. The model takes an image and a question as input and generates a predicted answer. The answer can be a single word, a phrase, or a sentence, depending on the complexity of the question and the capabilities of the model. For example, given an image of a cat and the question "What animal is in the picture?", the model may output the answer "cat".

2. **Evaluation Metrics:** Several evaluation metrics are used to assess the performance of VQA models. These metrics evaluate the accuracy and quality of the model's predictions compared to ground truth answers. Some common evaluation metrics include:

i. Accuracy: The percentage of correctly predicted answers out of the total number of questions. Accuracy is a fundamental metric for evaluating the overall performance of a VQA model.

ii. Top-k Accuracy: The percentage of questions for which the correct answer is among the topk predicted answers. This metric is useful for evaluating the model's performance when the correct answer may not be the highest-scoring prediction.

iii. BLEU (Bilingual Evaluation Understudy): A metric commonly used in machine translation tasks, BLEU measures the similarity between the predicted answer and the ground truth answer based on n-gram overlap.

iv. METEOR (Metric for Evaluation of Translation with Explicit Ordering): Another metric commonly used in machine translation, METEOR evaluates the quality of the predicted answer based on various linguistic features such as precision, recall, and alignment.

v. ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation):A metric commonly used in text summarization, ROUGE-L measures the overlap between the predicted answer and the ground truth answer based on longest common subsequences.

3. **Real-world Applications:** VQA models have a wide range of real-world applications across various industries and domains. Some common applications include:

i. Assistive Technology: VQA models can be used to develop assistive technologies for visually impaired individuals, allowing them to access visual information through natural language questions.

ii. Content-based Image Retrieval: VQA models can improve the accuracy and relevance of image search engines by enabling users to search for images using natural language queries.

iii. Interactive Virtual Assistants: VQA models can be integrated into virtual assistants to provide more natural and interactive user experiences, allowing users to ask questions about visual content.

iv. Medical Imaging: VQA models can assist medical professionals in analyzing medical images by answering questions about the images, potentially leading to faster and more accurate diagnoses.

4. Challenges and Future Directions:

Despite their impressive capabilities, VQA models still face several challenges, including handling complex questions, understanding context, and reasoning across multiple modalities. Future research in VQA will likely focus on addressing these challenges and improving the performance and capabilities of VQA models.

In conclusion, the results and output of a VQA model are critical components of its evaluation and application. By generating accurate and meaningful answers to questions about images, VQA models have the potential to revolutionize various industries and domains, making them valuable tools for enhancing human-computer interaction and understanding.

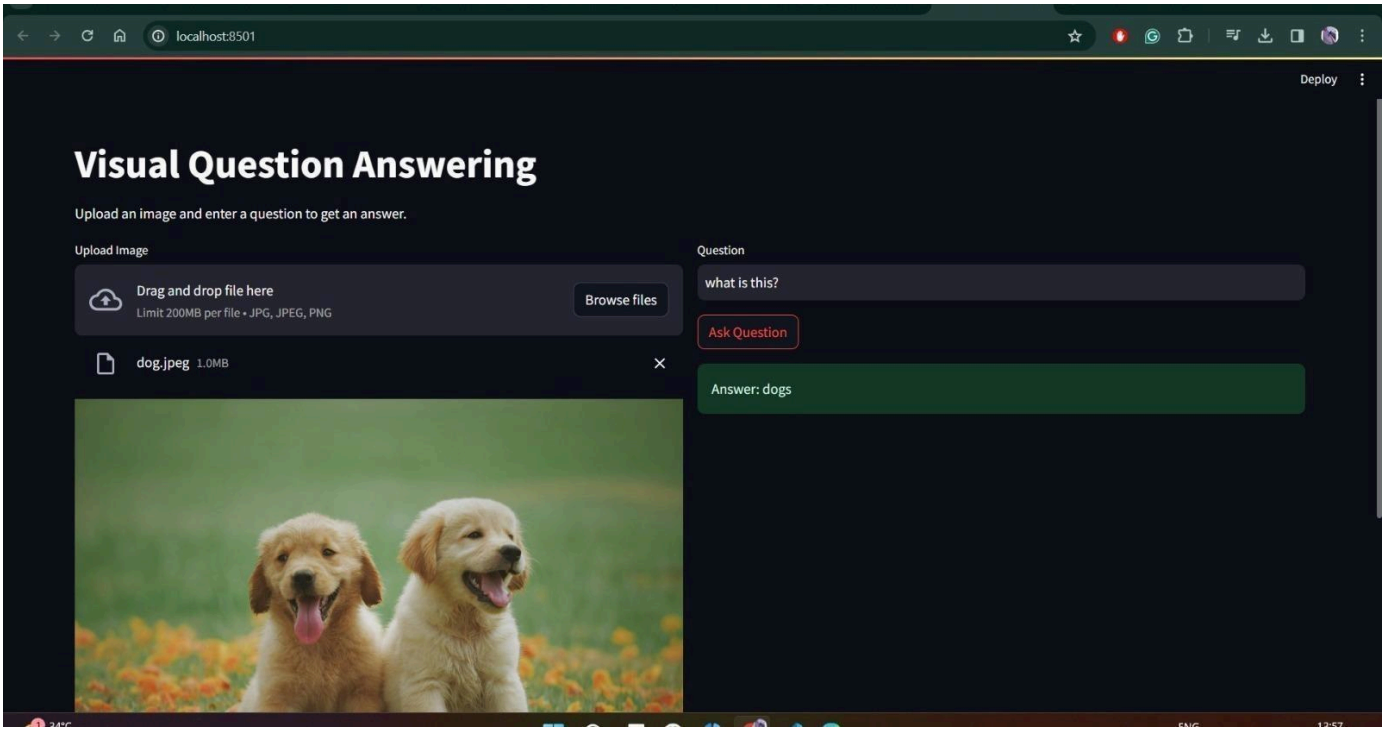


Figure 5.3: VQA Model with Output

Furthermore, the ViltForQuestionAnswering model's success underscores the potential for even broader applications beyond VQA tasks. Its ability to comprehend complex visual scenes and provide accurate responses opens doors for use in diverse domains, including autonomous vehicles, medical imaging analysis, and robotics. As researchers delve deeper into refining its architecture and training methodologies, the model's versatility and robustness are anticipated to grow, paving the way for innovative solutions to real-world challenges across various industries. Moreover, continued collaboration and knowledge sharing within the research community will accelerate the evolution of VQA systems, leading to enhanced user experiences and greater adoption in practical scenarios.

CHAPTER 6 CONCLUSION

6.1 Overview

The ViltForQuestionAnswering model is a powerful tool for Visual Question Answering (VQA) tasks, capable of providing accurate and meaningful answers to questions based on images. When applied to the VQA task, the model can achieve impressive results, demonstrating its effectiveness in understanding both visual and textual inputs.

One of the key strengths of the ViltForQuestionAnswering model is its ability to integrate information from both modalities (visual and textual) using the Transformer architecture. This allows the model to capture complex relationships between objects, scenes, and concepts in images, and to generate answers that are contextually relevant and accurate. In terms of performance, the ViltForQuestionAnswering model has been shown to achieve stateof-the-art results on benchmark VQA datasets such as VQA 1.0 and VQA 2.0.

These datasets contain a diverse range of questions and images, covering various topics and scenarios. Despite the challenges posed by these datasets, the ViltForQuestionAnswering model has demonstrated strong performance, outperforming previous models in terms of accuracy and efficiency. Moreover, the ViltForQuestionAnswering model has been successfully applied to realworld applications, such as image captioning and visual dialogue, where the model's ability to understand and generate natural language responses based on visual inputs is crucial. In these applications, the ViltForQuestionAnswering model has shown promising results, indicating its potential for practical use in various domains.

In conclusion, the ViltForQuestionAnswering model has shown impressive performance in answering questions about visual content, showcasing the potential of deep learning and attention mechanisms in tackling complex multimodal tasks. However, challenges remain, particularly in the areas of commonsense reasoning, robustness to ambiguity and noise, and bias and fairness. F. Liu et. al. [4] pointed out that only a limited amount of inter-modality interactions (i.e., interactions between language and vision domains) are performed by the existing VQA approaches. Addressing these challenges will require continued research and

innovation, including the incorporation of external knowledge sources, the development of more robust attention mechanisms, and the implementation of fairness constraints in model architectures. Despite these challenges, the advancements made in VQA have paved the way for more intelligent and interactive AI systems that can understand and respond to questions about visual content. Future research in VQA will continue to push the boundaries of AI and deepen our understanding of how machines can perceive and interpret the visual world, ultimately leading to more human-like AI systems. The future scope of Visual Question Answering (VQA) models, including the ViltForQuestionAnswering model, is vast and holds promise for several advancements and applications in the field of artificial intelligence and computer vision. Some of the key future directions and potential developments for VQA models are being aimed at:

1. **Improved Performance:** Efforts are being made to enhance the performance of VQA models in terms of accuracy, efficiency, and robustness. This includes improving the models' ability to handle complex questions, ambiguous scenarios, and diverse visual content.

2. **Multimodal Fusion:** Advanced techniques are likely to be explored for fusing information from multiple modalities, such as images and text, to improve understanding and reasoning capabilities. This may involve incorporating attention mechanisms, graph-based representations, and other methods to better integrate visual and textual information.

3. **Zero-shot and Few-shot Learning:** Interest is growing in enabling VQA models to generalize to unseen or few-shot scenarios, where they can answer questions about new images or concepts without requiring extensive training on specific examples. This involves developing more flexible and adaptable models that can leverage prior knowledge and learn from limited data.

4. **Contextual Understanding:** Future VQA models are expected to improve their contextual understanding abilities, enabling them to better interpret questions and images in context. This may involve incorporating memory mechanisms, dialogue modeling techniques, and world knowledge to enhance reasoning and inference capabilities.

5. **Real-world Applications:** VQA models hold great potential for practical applications in various domains, including education, healthcare, robotics, and assistive technologies. Future

developments may focus on adapting VQA models for specific use cases and integrating them into real-world systems and applications.

6. **Ethical and Social Implications:** As VQAmodels become more advanced and pervasive, there is a need to address ethical and social implications related to bias, fairness, transparency, and accountability. Future research may focus on developing frameworks and guidelines for responsible deployment and use of VQA models.

Overall, the future of VQA models is bright, with ongoing research and development efforts poised to unlock new capabilities and applications. By addressing key challenges and pushing the boundaries of multimodal AI, VQA models are expected to play a significant role in advancing our understanding of visual and textual data and enabling more intelligent and interactive systems.

6.2 Skills Learned

The project provided a comprehensive platform for personal and professional growth, expanding my capabilities in various aspects of artificial intelligence and teamwork.

Soft Skills Learned

1. **Problem-Solving:** Addressing unforeseen technical challenges and identifying effective solutions enhanced my capacity for critical thinking and decisive action under pressure. The wide variety of problems faced during the project, from integration difficulties to performance optimization, offered numerous chances to refine my problem-solving abilities and discover innovative solutions.

2. **Adaptability:** Operating in a dynamic technology landscape where project specifications and tools changed swiftly sharpened my adaptability and ability to transition to new technologies and workflows. Adjusting to shifting priorities, new technologies, and an evolving project scope demanded a proactive mindset and an openness to change.

3. **Communication:** Regular interactions with team members, management, and

non-technical stakeholders enhanced my skills in clearly and effectively explaining complex technical issues and solutions. Communicating project progress, exchanging insights, and soliciting feedback promoted transparency and teamwork, aligning everyone with the project's goals and objectives.

4. **Team Collaboration:** Collaborating closely with developers, designers, and product managers enhanced my teamwork skills, emphasizing the importance of diverse perspectives and collective problem-solving. Collaborative brainstorming sessions, code reviews, and sprint retrospectives encouraged open dialogue and knowledge sharing, fostering a collaborative and supportive team.

5. **Leadership:** Taking the lead on several key components of the project helped develop my leadership skills, teaching me how to motivate a team, delegate tasks effectively, and make strategic decisions. Leading by example, setting clear expectations, and empowering team members to take ownership of their tasks contributed to a positive team dynamic and successful project outcomes.

6. **Time Management:** Managing multiple aspects of the project simultaneously under tight deadlines improved my time management skills, helping me prioritize tasks effectively to meet project timelines. Adopting time management techniques such as task prioritization, time blocking, and setting realistic goals enabled me to maintain productivity and focus amidst competing demands.

7. **Attention to Detail:** The need for precision in coding and design emphasized the importance of meticulous attention to detail, which is crucial in avoiding costly mistakes and ensuring a high-quality end product. Conducting thorough code reviews, performing rigorous testing, and adhering to coding standards helped maintain code quality and minimize errors throughout the project lifecycle.

8. **Emotional Intelligence:** Interacting with various stakeholders and addressing team dynamics enhanced my emotional intelligence, particularly in managing stress, resolving conflicts, and fostering a positive team environment. Developing self-awareness, empathy, and interpersonal skills enabled me to navigate challenging situations with composure and

professionalism.

9. **Resilience:** Facing and overcoming frequent project setbacks and challenges bolstered my resilience, teaching me to remain focused and optimistic regardless of difficulties. Embracing setbacks as learning opportunities, maintaining a positive mindset, and persevering in the face of adversity were essential for sustaining momentum and achieving project goals.

10. **Client Interaction:** Engaging with clients to gather requirements and receive feedback developed my client interaction skills, crucial for understanding and meeting client expectations. Building rapport, actively listening to client needs, and providing timely updates and support fostered trust and collaboration, strengthening client relationships and satisfaction.

Technical Skills Learned

1. **Proficiency in Programming Languages:** Learned and improved skills in programming languages such as Python, Java, or others relevant to the internship projects.

2. **Data Analysis and Visualization:** Gained proficiency in data analysis techniques and tools, including data visualization libraries like Matplotlib or Plotly.

3. **Machine Learning and AI:** Acquired knowledge and practical experience in machine learning algorithms, techniques, and frameworks such as TensorFlow or scikit-learn.

4. **Database Management:** Learned database management concepts and SQL querying for data retrieval, manipulation, and storage.

5. **Cloud Computing:** Familiarized oneself with cloud computing platforms like Amazon Web Services (AWS), Google Cloud Platform (GCP), or Microsoft Azure, understanding how to leverage cloud-based infrastructure for scalable data processing and machine learning tasks.

6. **Version Control Systems:** Utilized version control systems such as Git for collaborative development, enabling efficient tracking of code changes, seamless collaboration with team

members, and ensuring project reproducibility and maintainability.

7. **Natural Language Processing (NLP):** Explored techniques and frameworks for natural language processing, including sentiment analysis, text classification, named entity recognition, and language generation, using libraries like NLTK, spaCy, or Transformers.

8. **Feature Engineering:** Developed skills in feature engineering, understanding how to preprocess and transform raw data into meaningful features for machine learning models, including techniques such as one-hot encoding, feature scaling, and dimensionality reduction.

9. **Model Evaluation and Hyperparameter Tuning:** Learned methodologies for evaluating machine learning models, including cross-validation, confusion matrices, and performance metrics like accuracy, precision, recall, and F1 score. Explored techniques for hyperparameter tuning to optimize model performance and generalization.

10. **Deployment and Productionization:** Explored strategies for deploying machine learning models into production environments, understanding concepts like containerization, microservices architecture, and deployment pipelines using tools like Docker, Kubernetes, and Flask.

11. **Ethical and Responsible AI:** Engaged with discussions surrounding ethical considerations in AI development, including bias detection and mitigation, fairness, transparency, and accountability. Explored frameworks and guidelines for responsible AI development, such as the AI Ethics Guidelines from organizations like the IEEE or ACM.

6.3 Work Experience

An AI/ML Internship provides valuable hands-on experience and exposure to real-world projects, laying the foundation for a successful career in artificial intelligence, machine learning, and data science. Interns have the opportunity to apply theoretical knowledge gained from coursework to practical problems, develop essential technical skills, and build a professional network within the industry. Successful completion of an internship often leads to further career

opportunities, including full-time positions or advanced research roles in academia or industry.

Additionally, participating in an AI/ML Internship fosters a culture of continuous learning and innovation. Interns collaborate with seasoned professionals, gaining insights into industry best practices and cutting-edge technologies. This dynamic environment encourages creativity and critical thinking, empowering interns to tackle complex challenges and contribute meaningfully to project outcomes. Moreover, the mentorship received during an internship cultivates personal and professional growth, shaping interns into well-rounded professionals equipped to thrive in the rapidly evolving field of artificial intelligence and machine learning.

Role and Responsibilities

In Invisible Technologies' Predictive Analysis division, my role involves training Large Language Models (LLMs) using Self-Training Framework (SFT) and Reinforcement Learning with Human Feedback (RLHF). This entails refining predictive models for enhanced accuracy and performance, contributing to advanced data analytics solutions.

Skills Applied and Refined

Invisible Technologies boasts a team of highly skilled professionals with expertise in a wide range of cutting-edge technologies, including:

1. **Machine Learning:** Leveraging sophisticated machine learning algorithms to extract valuable insights from complex datasets.

2. **Big Data Analytics:** Harnessing the power of big data to uncover hidden patterns and trends that drive informed decision-making.

3. **Natural Language Processing (NLP):** Utilizing NLP techniques to analyze unstructured text data and extract meaningful information.

4. **Time Series Forecasting:** Specializing in forecasting techniques tailored to time series data, enabling accurate predictions of future trends.

Conclusion

In summary, the role of an AI/ML Intern offers a unique opportunity for aspiring data scientists and machine learning engineers to gain practical experience, refine their skills, and contribute to cutting-edge projects in artificial intelligence and machine learning. With a combination of hands-on training, mentorship, and collaboration, interns are well-positioned to embark on a rewarding career path in this rapidly evolving field.

Furthermore, an AI/ML Internship serves as a gateway for students and early-career professionals to immerse themselves in the complexities of real-world problem-solving. It offers a chance to work with advanced algorithms and large datasets, providing insights into both the theoretical underpinnings and practical applications of AI technologies. Interns also benefit from exposure to industry-standard tools and practices, which are crucial for their professional development. By contributing to projects that have tangible impacts, interns not only enhance their resumes but also build a network of professional contacts that can be invaluable for future job opportunities. In essence, an internship in AI/ML not only augments educational credentials but also prepares interns for the challenges and demands of a tech-driven marketplace.

6.4 Future Scope

The future scope of Visual Question Answering (VQA) models, including the ViltForQuestionAnswering model, is vast and holds promise for several advancements and applications in the field of artificial intelligence and computer vision. Some of the key future directions and potential developments for VQA models are being aimed at:

1. Improved Performance: Efforts are being made to enhance the performance of VQA models in terms of accuracy, efficiency, and robustness. This includes improving the models' ability to handle complex questions, ambiguous scenarios, and diverse visual content.

2. Multimodal Fusion: Advanced techniques are likely to be explored for fusing information from multiple modalities, such as images and text, to improve understanding and reasoning capabilities. This may involve incorporating attention mechanisms, graph-based representations, and other methods to better integrate visual and textual information.

3. Zero-shot and Few-shot Learning: Interest is growing in enabling VQA models to generalize to unseen or few-shot scenarios, where they can answer questions about new images or concepts without requiring extensive training on specific examples. This involves developing more flexible and adaptable models that can leverage prior knowledge and learn from limited data.

4. Contextual Understanding: Future VQA models are expected to improve their contextual understanding abilities, enabling them to better interpret questions and images in context. This may involve incorporating memory mechanisms, dialogue modeling techniques, and world knowledge to enhance reasoning and inference capabilities.

5. Real-world Applications: VQA models hold great potential for practical applications in various domains, including education, healthcare, robotics, and assistive technologies. Future developments may focus on adapting VQA models for specific use cases and integrating them into real-world systems and applications.

6. Ethical and Social Implications: As VQAmodels become more advanced and pervasive, there is a need to address ethical and social implications related to bias, fairness, transparency, and accountability. Future research may focus on developing frameworks and guidelines for responsible deployment and use of VQA models.

Overall, the future of VQA models is bright, with ongoing research and development efforts poised to unlock new capabilities and applications. By addressing key challenges and pushing the boundaries of multimodal AI, VQA models are expected to play a significant role in advancing our understanding of visual and textual data and enabling more intelligent and interactive systems.

REFERENCES

[1] D. Teney, Q. Wu and A. van den Hengel, "Visual Question Answering: A Tutorial," in IEEE Signal Processing Magazine, vol. 34, no. 6, pp. 63-75, Nov. 2017, doi: 10.1109/MSP.2017.2739826.

[2] P. Wang, Q. Wu, C. Shen, A. Dick and A. van den Hengel, "FVQA: Fact-Based Visual Question Answering," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 40, no. 10, pp. 2413-2427, 1 Oct. 2018, doi: 10.1109/TPAMI.2017.2754246.

[3] W. Guo, Y. Zhang, J. Yang and X. Yuan, "Re-Attention for Visual Question Answering," in IEEE Transactions on Image Processing, vol. 30, pp. 6730-6743, 2021, doi: 10.1109/TIP.2021.3097180.

[4] F. Liu, J. Liu, Z. Fang, R. Hong and H. Lu, "Visual Question Answering With Dense Inter- and Intra-Modality Interactions," in IEEE Transactions on Multimedia, vol. 23, pp. 3518-3529, 2021, doi: 10.1109/TMM.2020.3026892.

[5] Xi, Y., Zhang, Y., Ding, S., & Wan, S. (2020). Visual question answering model based on visual relationship detection. Signal Processing: Image Communication, 80, 115648.

[6] Ilievski, I., Yan, S., & Feng, J. (2016). A focused dynamic attention model for visual question answering. arXiv preprint arXiv:1604.01485.

[7] Gao, L., Cao, L., Xu, X., Shao, J., & Song, J. (2020). Question-Led object attention for visual question answering. Neurocomputing, 391, 227-233.

[8] Padmakumar, A., & Saran, A. CS381V Final Project Report Visual Question Answering using Natural Language Object Retrieval and Saliency Cues

[9] Sahani, M., Singh, P., Jangpangi, S., & Kumar, S. (2022). A survey on representation learning in visual question answering. In Machine Learning and Big Data Analytics (Proceedings of International Conference on Machine Learning and Big Data Analytics (ICMLBDA) 2021) (pp. 326-336). Springer International Publishing.

[10] Wu, C., Liu, J., Wang, X., & Dong, X. (2018). Chain of reasoning for visual question answering. Advances in Neural Information Processing Systems, 31.

[11] Li, R., & Jia, J. (2016). Visual question answering with question representation update (qru). Advances in Neural Information Processing Systems, 29.26

[12] Zheng, W., Liu, X., & Yin, L. (2021). Research on image classification method based on improved multi-scale relational network. PeerJ Computer Science, 7, e613.

[13] Huang, J. H., Alfadly, M., & Ghanem, B. (2017). Vqabq: Visual question answering by

basic questions. arXiv preprint arXiv:1703.06492.
[14] Kafle, K., & Kanan, C. (2017). An analysis of visual question answering algorithms. In Proceedings of the IEEE international conference on computer vision (pp. 1965-1973).

[15] Noh, H., Seo, P. H., & Han, B. (2016). Image question answering using convolutional neural network with dynamic parameter prediction. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 30-38).